INFORMATION SHEET 6

SOIL MANAGEMENT

Soil is formed from decomposing rock. Depending on the type of rock, different soils have developed over the centuries and are still developing. The structure and characteristics of soils and their capacity to support plant growth differ according to the size of the soil particles and the content of organic and inorganic matter.

COMMON SOIL TYPES

Some types of soil are more fertile than others. River plains and soils of volcanic origin are naturally fertile. Some soils can be acidic. Common soil types found in Africa, their features, possible improvements and some of their management aspects are described in Table 1.

TABLE 1
Common soil types and treatments

Soil type	Features	Methods of improvement
Sand	• Poor structure • Poor fertility • Cannot hold water	• Add organic matter and fertilizer regularly • Use green manure crops • Add anthill material • Practise minimum tillage
Silt	• Poor structure	• Add coarse organic matter
Clay	• Dries hard Holds too much water	• Add organic matter, compost and gypsum*
Acid subsoil	• Subsoil layer is toxic to some plants	• Grow shallow-rooted plants (vegetables) Apply ground limestone (based on soil analysis results) and manure
Loam	• Mixture of sand, silt and clay	• Maintain soil fertility with periodic application of fertilizer and compost

* Mineral consisting of hydrated calcium sulphate that occurs in sedimentary rocks and clay and is used in the making of plaster, cement, etc.

If the soil is kept in good condition - protected and "fed" with the nutrients that the plants require - the home garden can be cultivated year round and year after year.

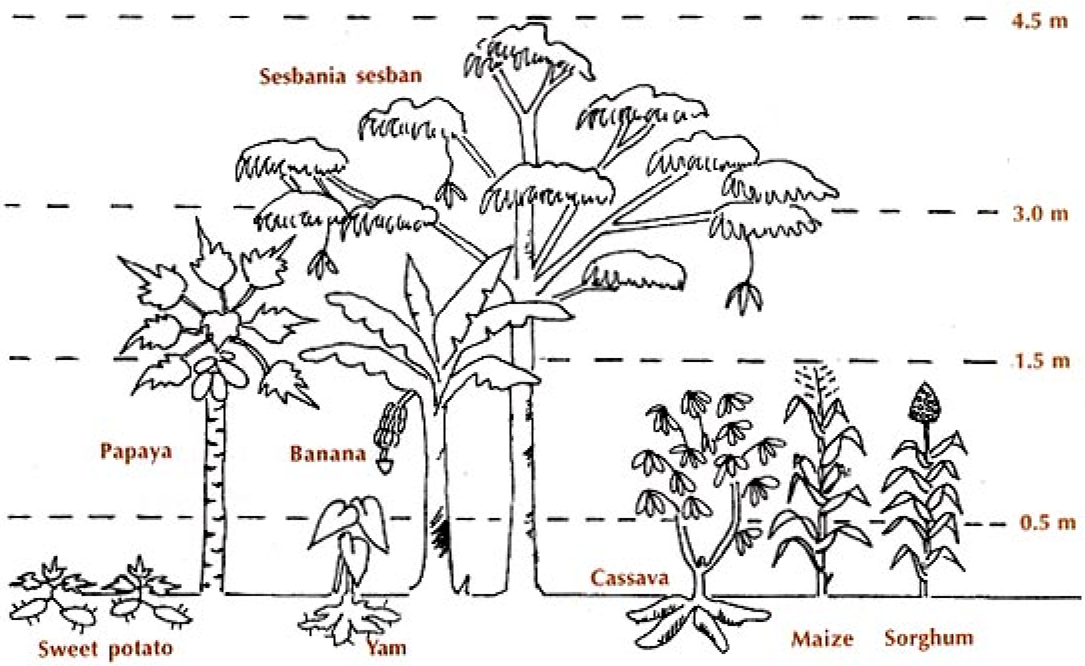
SOIL EROSION

The first step in soil management is preventing the loss, or erosion, of soil. Topsoil is particularly vulnerable to erosion if not protected by plants or mulch or by other measures. The soil that remains after the loss of topsoil is usually less productive, which results in lower production from home garden crops. The challenge is to protect home garden soil while using the land for food production and other non-food activities.

Soil erosion is caused mainly by wind and water but also by incorrect cultivation practices. Rain and wind dislodge and then carry away soil particles. Where the soil is bare or the vegetation poor, rainwater does not seep into the soil; instead it runs off and carries with it loose topsoil. Sloping land and light soils with low organic matter content are both prone to erosion. Once eroded, the soil is lost forever.

Soil erosion is a problem in regions with little vegetation, particularly in the semi-arid and arid zones of Africa. In the humid tropics, erosion was not considered a problem when the land was in its natural state, because the variety of native plants kept the soils covered at all times. Now, people are clearing more land for agricultural purposes, and the situation has changed. Heavy rains coupled with poor soil management of cultivated areas are now common causes of soil erosion in the humid areas.

FIGURE 1
Plants of different heights protect the soil



Water erosion

There are three common forms of water erosion.

- Sheet erosion: a thin top layer of soil is removed from the soil by the impact of rain. With sheet erosion, small heaps of loose material (e.g. grass) amass between fine lines of sand after a rainstorm. This erosion takes place across a whole garden or field.
- Rill erosion: water flows over minor depressions on the land's surface and cuts small channels into the soil. The erosion takes place along the length of these channels.
- Gully erosion: a gully forms along natural depressions on the soil's surface or on slopes. The head of a gully moves up the slope in the opposite direction of the flow of water. Gullies are symptoms of severe erosion.

Wind erosion

This occurs mostly on light soils and bare land. High winds cause severe damage. Wind erosion is a common problem in dry and semi-arid areas, as well as in areas that get seasonal rains.

Unlike water which only erodes on slopes, wind can remove soil from flat land as well as from sloping land; it can also transport the soil particles through the air and deposit them far away. Soils vulnerable to wind erosion are dry, loose, light soils with little or no vegetative cover.

Ploughing up and down a slope causes soil erosion. To prevent the loss of home garden soils, certain measures must be taken. These include:

- clearing only the land to be cultivated;
- planting along a contour[8] and using grassed channels;
- establishing windbreaks and bench terraces;
- ploughing along a contour;
- planting cover crops and mulching.

When clearing land for cultivation, the beneficial effects of certain trees and plants should be considered. Some trees should be left, since they may supply food, medicine, shade or, when they shed their leaves, organic matter. Information on how to establish contour lines is given in Home Garden Technology Leaflet 7 "Erosion control and soil conservation".

FEEDING THE SOIL

One of the main goals of home garden development is to make the garden's soil fertile and well structured, so a wide range of useful crops can grow and produce well. In order to grow, plants require nutrients that are present in organic matter, such as *nitrogen, calcium* and *phosphorus*, as well as *minerals* and *trace elements*.

If the natural fertility or structure of the soil is poor, it must be continuously "fed" with organic matter, such as leaves and manure, in order to improve its productivity and water-holding capacity. As organic matter decomposes, it becomes food for plants. It also improves soil structure by loosening heavy clay and binding sandy soil.

Feeding the soil with organic matter is especially important in the early years of home garden development. Organic matter (i.e. waste from plants and livestock) can be collected and buried in the soil, where it will decompose. The organic matter also can be used to make *compost*, which can be applied to the soil to enrich its fertility.

The roots of legumes contain nitrogen-fixing bacteria. Therefore, intercropping or rotating legumes with other crops helps maintain or improve the nitrogen content of the soil, and this enhances the growth of other plants.

Healthy plants yield more and are better protected from insects and disease. The application of organic matter, such as compost, animal manure, green manure and soil from anthills, improves soil structure and adds nutrients to the soil.

LONG-TERM SOIL MANAGEMENT

The ideal way to protect and feed the soil is to apply organic matter or compost regularly and to keep the soil covered with plants. A multilayer cropping system in which a mixture of trees and other plants with different maturity times are grown together will protect the soil and recycle nutrients. Leguminous plants such as cowpeas, groundnuts and beans are particularly useful in providing continuous nutrients for garden crops.

Further soil management information for home garden managers is included in Home Garden Technology Leaflets 5, "Soil improvement", 6, "Special techniques for improving soil and water management", 7, "Erosion control and soil conservation", 8, "Use of sloping land", and 9, "Cover cropping".

---

[8] Contour lines trace horizontally across the slope, joining points of the same elevation.